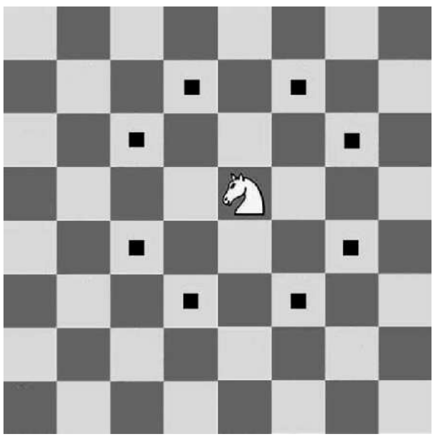
G. The Knights of Königsberg

time limit per test: 1 second

memory limit per test: 256 megabytes

You are given a chessboard of size $N \times M$. There are K knights placed on distinct cells of this board. The standard movement of a knight in chess is as follows: it moves in an "L" shape —either two squares in one direction and then one square perpendicular to that, or one square in one direction and then two squares perpendicular to that. A knight can jump over other pieces.



Your task is to determine whether any pair of knights are attacking each other. If at least one pair of knights are in positions such that one can attack the other in a single move, print "YES". Otherwise, print "NO".

Input

The first line contains three integers N, M, and K $(1 \leq N, M \leq 1000, 0 \leq K \leq 10^5)$ — the number of rows, the number of columns, and the number of knights on the board.

Each of the next K lines contains two integers x_i, y_i $(1 \leq x_i \leq N, 1 \leq y_i \leq M)$ — the position of the i-th knight on the board.

It is guaranteed that all K knights are placed on distinct positions.

Examples

input Copy

```
6 4 4
2 3
4 2
1 2
3 3
```

output Copy

```
YES
```

input Copy

```
4 4 4
1 1
1 3
2 2
2 4
```

CSE221 Assignment 04 Summer 2026

Contest is running

17:11:23

Contestant

→ Submit?

Language: Java 21 64bit

Choose file: Choose File No file chosen

Success!

CLOUDFLARE

Privacy • Help

Submit

output
NO